Geog 4/6300: Lab 2

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Spatial statistics with vector and raster data

Overview:

In this lab, you'll work with point, polygon, and raster data to investigate the distribution and environmental characteristics of trees in Georgia.

You'll begin with observations of two tree species collected through the iNaturalist citizen-science platform. First, you'll use these observations to calculate the number of trees of each species found in Georgia's counties. You'll then use these counts to calculate a **location quotient (LQ)**, a measure that compares the prevalence of a species in each county with its prevalence across the state as a whole.

In the second part of the lab, you'll use temperature data from the ERA5 climate reanalysis to investigate the temperatures associated with observations of the two tree species. You'll extract temperature values from a raster for each tree observation and then calculate descriptive statistics for each species.

The ERA5 data we are using were obtained through the Google Earth Engine platform. More information is available at https://github.com/communitymaplab/era5_1950-69_2014-2024.

For this lab to be marked complete, the following criteria must be met:

- Your responses should show your ability to load and manipulate point and polygon vector data. (Tasks 1–3)
- Your responses should show your ability to calculate counts and create new variables from spatial data. (Tasks 2–4)
- Your responses should show your understanding of the location quotient as a measure of spatial concentration. (Tasks 4–5)
- Your responses should show your ability to load raster data and extract raster values at point locations. (Tasks 6–8)
- Your responses should show your ability to manipulate data frames and calculate summary statistics. (Tasks 9–10)
- Your responses should demonstrate your ability to describe statistical and spatial patterns in geographic data.

Before you get started, you'll need to load the following packages: `tidyverse`, `terra`, `sf`, and `tmap`. You may also want to use `knitr` for displaying tables.

```
# Load packages here.

library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.1     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ ggplot2   4.0.3     ✓ tibble    3.3.1
✓ lubridate 1.9.5     ✓ tidyr     1.3.2
✓ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

```
library(terra)
```

```
terra 1.9.46

Attaching package: 'terra'

The following object is masked from 'package:tidyr':

    extract
```

```
library(sf)
```

```
Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(tmap)
library(knitr)
```

```
Attaching package: 'knitr'

The following object is masked from 'package:terra':

    spin
```

Part 1: Mapping the distribution of trees

Where are these tree species most prevalent in Georgia?

In this part of the lab, you'll use iNaturalist observations of two tree species to investigate their geographic distribution.

The data folder contains a file named `gatree_obs_inat.csv`. This file contains observations of live oaks, pecan trees, flowering dogwoods, and great rhododendrons in Georgia.

Your first task is to load these observations and convert them into a spatial point dataset. The observations contain longitude and latitude coordinates that you can use to create the point locations.

Task 1

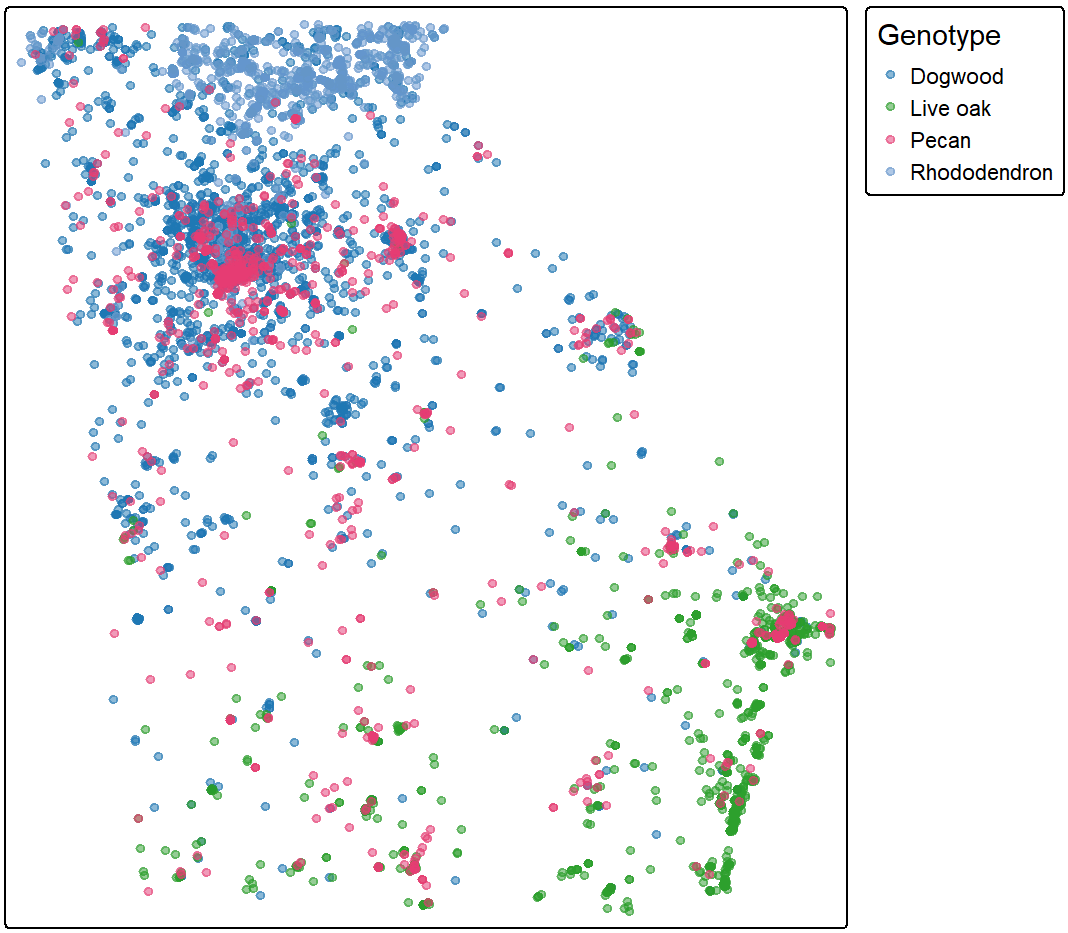
Load the iNaturalist observations and convert them to an `sf` point dataset using the longitude and latitude fields. Map the observations using `tmap`, with the point colors based on the `gentype` field.

```
gatree_obs_inat <- read_csv("data/gatree_obs_inat.csv") |>
  st_as_sf(coords = c("longitude", "latitude"),
           crs = 4326,
           remove = FALSE)
```

```
Rows: 6322 Columns: 19
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (13): uuid, observed_on_string, time_observed_at, time_zone, user_login...
dbl   (5): id, user_id, taxon_id, longitude, latitude
date  (1): observed_on

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
tm_shape(gatree_obs_inat) +
  tm_dots(size = 0.3, fill_alpha = 0.5,
  fill = "gentype",
  fill.legend = tm_legend("Genotype"))
```



Now that you have mapped the observations, you'll use the Georgia county boundaries to determine how many observations of each species occur within each county.

Load the county data from:

`ACSCtyData_2022ACS_simplify.gpkg`

You'll then use a spatial join to determine which county contains each tree observation.

Task 2

Load the Georgia county dataset and use a spatial join to assign each tree observation to a county. Create a table showing the number of observations of each species in each county.

```
ga_cty <- st_read("data/ACSCtyData_2022ACS_simplify.gpkg") %>%
  st_transform(4326) %>%
  filter(st_name == "Georgia")
```

```
Reading layer `ACSCtyData_2022ACS_simplify' from data source
  `C:\Users\wbm11\OneDrive\Documents\GitHub\William_Burgett-
Merkel_GEOG6300_Lab2\data\ACSCtyData_2022ACS_simplify.gpkg'
  using driver `GPKG'
```

```
Simple feature collection with 3107 features and 65 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -124.7322 ymin: 24.54423 xmax: -66.96015 ymax: 49.38437
Geodetic CRS:  WGS 84
```

```
gatree_county<-gatree_obs_inat %>%
  st_join(ga_cty %>% st_transform(4326),
          join=st_within)

gatree_county_count<-gatree_county %>%
  count(cty_name, gentype)

gatree_county_count_wide<-gatree_county_count %>%
   st_drop_geometry() |>
  pivot_wider(names_from = gentype,
              values_from = n)
```

You should now have a dataset showing the number of observations of each species in each county.

For example, your table might have a structure similar to:

County	Live oak	Great rhododendron
Clarke	25	8
Oconee	17	12
...	...	...

Your next step is to calculate the **total number of tree observations** in each county.

Task 3

Create a county-level dataset containing the number of observations of each species and the total number of tree observations.

```
gatree_county_count<-gatree_county %>%
  count(cty_name, gentype)

gatree_county_count_wide<-gatree_county_count %>%
   st_drop_geometry() |>
  pivot_wider(names_from = gentype,
              values_from = n)

gatree_county_count_all<-gatree_county_count_wide |>
  group_by(cty_name) |>
  summarize(total_observations
            =sum(c(Dogwood, `Live oak`, Pecan, Rhododendron), na.rm = TRUE))
```

```
gatree_county_count_task3<-gatree_county_count_wide |>
  full_join(gatree_county_count_all, by = "cty_name")
```

Calculating a location quotient

You will now use these counts to calculate a **location quotient (LQ)**. For this lab, we'll use it to ask:

> **Is a particular tree species more prevalent among tree observations in a county than it is across Georgia as a whole?**

For this lab, calculate an LQ for **live oak observations**.

First, calculate the statewide proportion of observations that are live oak. Then calculate the proportion of observations that are live oak in each county. Finally, divide the county proportion by the statewide proportion.

Task 4

Calculate the location quotient for live oak observations for every Georgia county. Add the resulting LQ as a new variable in your county dataset.

```
gatree_county_count_pct_subarea<-gatree_county_count_task3 |>
  mutate(pct_subarea = `Live oak`/total_observations)

gatree_county_count_all_oak<- sum(gatree_county_count_task3$`Live oak`, na.rm = TRUE)
gatree_county_count_total_observation_count<-sum(gatree_county_count_task3$total_observations)

pct_area = gatree_county_count_all_oak/gatree_county_count_total_observation_count

gatree_county_count_task4<-gatree_county_count_pct_subarea |>
  mutate(Location_Quotient=pct_subarea/pct_area)
```

Now map your LQ values.

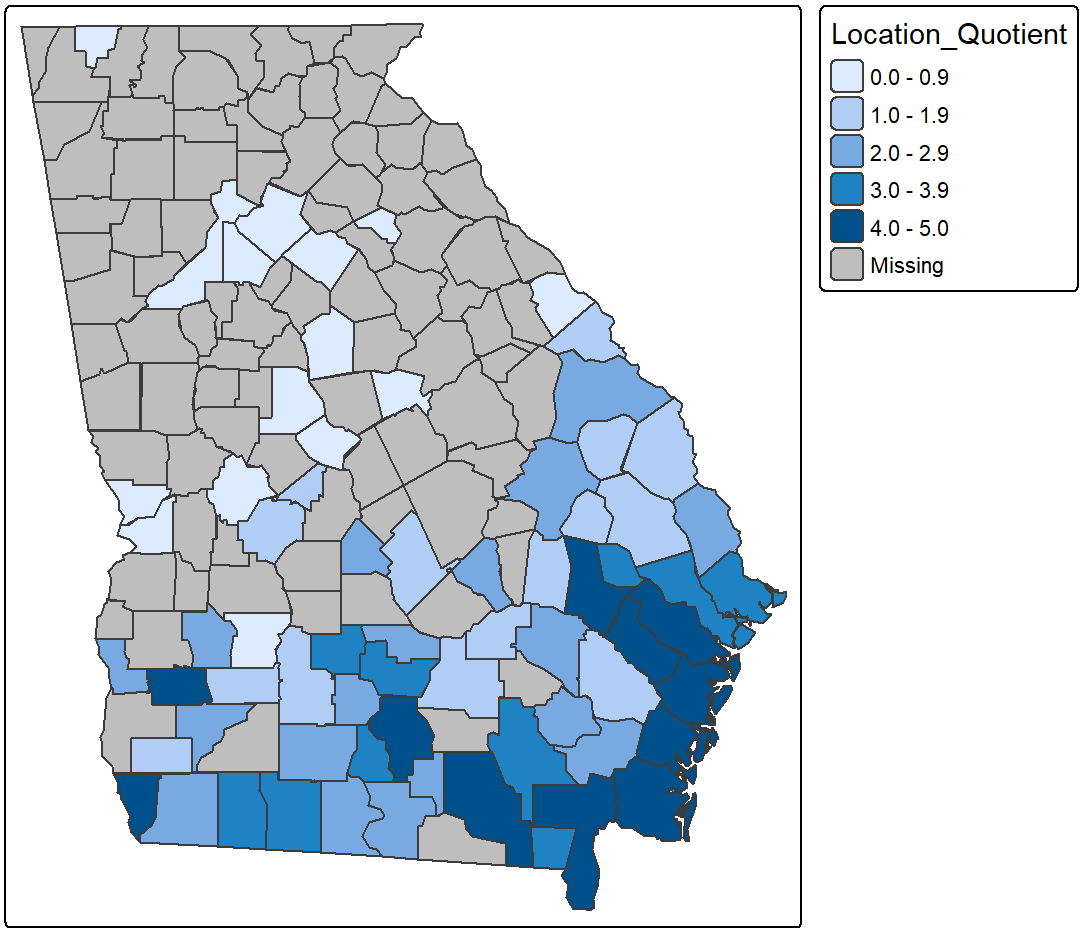
Task 5

Create a map showing the location quotient for live oak observations. Describe the geographic pattern you see. Which parts of Georgia have relatively high concentrations of live oak observations? Which have relatively low concentrations?

```
ga_cty_geom<-ga_cty |>
  select(cty_name, geom)

gatree_county_count_task5<-ga_cty_geom |>
  left_join(st_drop_geometry(gatree_county_count_task4),
            by = "cty_name")
```

```
tm_shape(gatree_county_count_task5)+
  tm_polygons("Location_Quotient")
```



Your response:

The Live Oaks' geographic pattern shows that they prefer southern Georgia, particularly along coastal Georgia. There are slight hot spots in counties with larger populations like the Atlanta metro area, Athens-Clarke County, and Macon, Ga, but they generally tend to be in the south, close to the coast, where they are most concentrated. Alternatively, they are not generally concentrated in North Georgia, and are not usually located in more rural areas north of Atlanta.

Part 2: Trees and temperature in Georgia

What temperatures are associated with observations of each species?

In the second part of the lab, you'll investigate the environmental characteristics of the four tree species.

The temperature data you'll use come from the ERA5 climate reanalysis. The data provide retrospective information about atmospheric conditions across the globe.

You will use the July temperature rasters from 2022–2024 to calculate a mean July temperature raster. You'll then extract the temperature associated with each tree observation.

Your first step is to load one of the temperature rasters—the one for July 1, 2024—and convert the temperature from degrees Kelvin to degrees Celsius.

You should see this raster in the data folder of this lab repository. Load it using the `rast` function from `terra`.

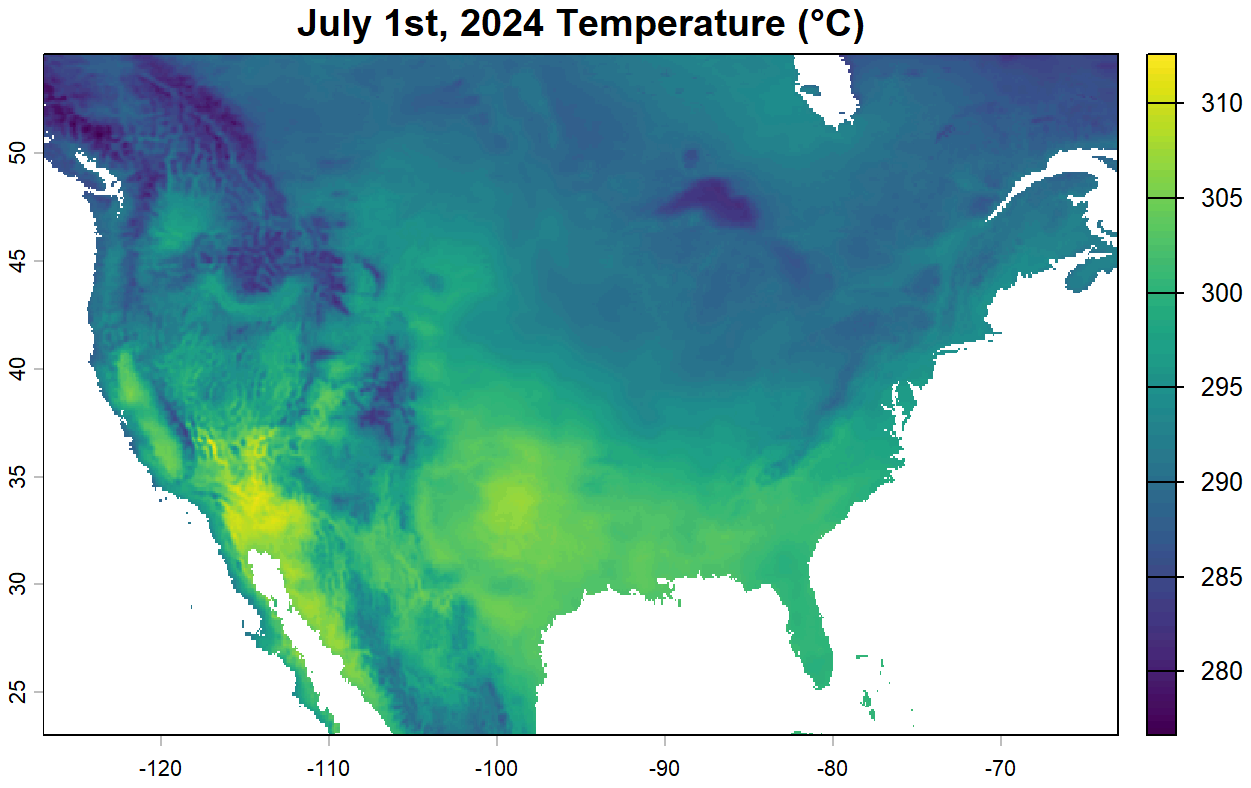
Task 6

Load the July 1, 2024 temperature raster and convert it from Kelvin to Celsius. Plot it when you are done.

```
temp_07012024<-rast("data/temp_july2022_2024/temperature_2m_2024-07-01.tif")

temp_07012024_C<-temp_07012024 - 273.15

plot(temp_07012024, main = "July 1st, 2024 Temperature (°C)")
```



Now load **all of the temperature rasters for July 2022–2024**.

Create a list of the files in the `temp_july2022_2024` folder and load them using `rast`. Convert the values from Kelvin to Celsius.

Use `nlyr()` to confirm how many raster layers you have loaded.

Task 7

Load all temperature rasters for July dates in 2022–2024, convert them from Kelvin to Celsius, and calculate the mean temperature across all of the raster layers.

```
temp2022_to_2024_files<-list.files("data/temp_july2022_2024",
                     pattern=".tif",
                     full.names=T)

temp_2022_to_2024<-rast(temp2022_to_2024_files)

nlyr(temp_2022_to_2024)
```

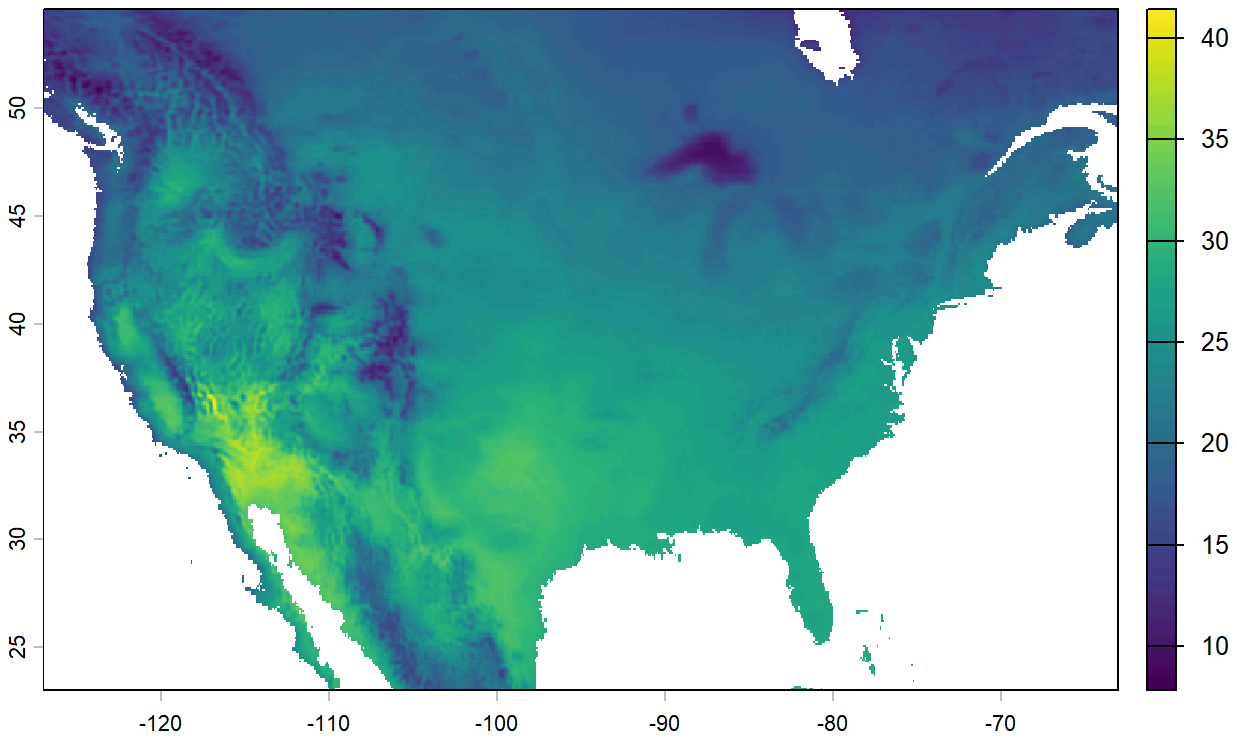
```
[1] 92
```

```
temp_2022_to_2024_C<-temp_2022_to_2024 - 273.15

mean(temp_2022_to_2024_C)
```

```
class       : SpatRaster
size        : 352, 713, 1  (nrow, ncol, nlyr)
resolution  : 0.08983153, 0.08983153  (x, y)
extent      : -127.0218, -62.9719, 22.99687, 54.61757  (xmin, xmax, ymin, ymax)
coord. ref. : lon/lat WGS 84 (EPSG:4326)
source(s)   : memory
name        :      mean
min value   :  7.824192
max value   : 41.435223
```

```
plot(mean(temp_2022_to_2024_C))
```



You now have a raster representing the mean July temperature across the study period.

Next, use `terra::extract()` to obtain the temperature value at the location of every tree observation.

Task 8

Extract the mean July temperature for each tree observation and add the temperature value to the tree dataset. Display the first several observations using `kable()`.

```
extracted_tree_temp_data_2022_to_2024<-terra::extract(
  x = mean(temp_2022_to_2024_C),
  y = gatree_obs_inat)

gatree_obs_inat$Temperature_2022_to_2024_C<-extracted_tree_temp_data_2022_to_2024$mean

kable(head(gatree_obs_inat))
```

id	uuid	observed_on_string	observed_on	time_observed_at	time_zone	user_id	user_login	use
38491	2be14d3a-d5c3-4d59-9a64-8ed5376f119f	November 7, 2011 6:29:23 PM EST	2011-11-07	2011-11-07 23:29:23 UTC	Eastern Time (US & Canada)	935	petermayfield	NA

id	uuid	observed_on_string	observed_on	time_observed_at	time_zone	user_id	user_login	use
145616	bc4be566-7fb3-43a8-a672-01900fa1a402	Fri Nov 09 2012 10:50:51 GMT-0500 (EST)	2012-11-09	2012-11-09 15:50:51 UTC	Eastern Time (US & Canada)	2179	charlie	natu cha
399515	2b86ad80-77d9-4ce6-8965-546a82a69722	2013-09-10	2013-09-10	NA	Eastern Time (US & Canada)	22176	lauralaneir	NA
618291	d3b145ff-3140-4d50-be18-66718b232a21	Mon Apr 07 2014 16:05:11 GMT-0400 (EDT)	2014-04-07	2014-04-07 20:05:11 UTC	Eastern Time (US & Canada)	27509	richard_candler	Rich Can
710549	9ba7a4c5-e68e-4b9c-94f7-792f8860f259	2013-10-13 06:21:49	2013-10-13	2013-10-13 10:21:49 UTC	Eastern Time (US & Canada)	379	thall	Trac
751374	fd172542-62b7-462d-bd96-43cee9cdfa11	2014-04-06	2014-04-06	NA	Eastern Time (US & Canada)	19847	carolcummings	NA

You can now calculate summary statistics for the two tree species.

For each species, calculate:

- the mean temperature
- the standard deviation of temperature
- the mean longitude
- the mean latitude

The mean longitude and latitude provide a simple measure of the **mean center** of the observations.

Task 9

Calculate the mean and standard deviation of temperature for each species, along with the mean longitude and latitude of their observations. Display the results using `kable()`.

```
Tree_Summary_Statistics_2022_to_2024<-gatree_obs_inat |>
group_by(gentype) |>
  summarize(
    mean_temp = mean(Temperature_2022_to_2024_C, na.rm = TRUE),
    standard_deviation = sd(Temperature_2022_to_2024_C, na.rm = TRUE),
    mean_longitude = mean(longitude, na.rm = TRUE),
    mean_latitude = mean(latitude, na.rm = TRUE)
    )
```

```
kable(Tree_Summary_Statistics_2022_to_2024)
```

gentype	mean_temp	standard_deviation	mean_longitude	mean_latitude	geometry
Dogwood	26.26175	0.8697368	-84.16345	33.81103	MULTIPOINT ((-82.80754 34.3…
Live oak	27.86903	0.2431182	-81.73671	31.63798	MULTIPOINT ((-85.2619 34.90…
Pecan	26.77629	0.7109518	-83.70809	33.23867	MULTIPOINT ((-82.86271 34.3…
Rhododendron	24.37448	1.0678707	-83.98433	34.66254	MULTIPOINT ((-83.58346 34.5…

Task 10

What does the table tell you about the temperatures associated with observations of the two species? How do their geographic distributions compare? Are there any similarities or differences that you find interesting?

First, to clarify, I couldn't find anything about 2 species, but rather 4, so I went ahead with 4 species. The table tells us the climate each tree likes to live in, and what middle point of all the trees is. The table also tells us that the Live Oaks live primarily in Southern Georgia, particularly along the coast as the mean center was locateda bit southwest of Savannah, GA. The other extreme was the Rhododendron, whichprimarily lived at the edge of Georgia/Tennessee border. The Pecan and Dogwood mean center were located essentially in the middle of the state, which could mean it is concentrated in the middle of the state, or is widespread throughoutthe state. A kernel density or location quotient could tell which case is true. I think the difference between the Rhododendron and the Live Oak is pretty interesting, because they are on complete opposite ends of the state, which means they both prefer very different climates (warm and moist for Live Oak, higher elevation, dryer and cooler for the Rhododendron).

Lab challenge

The data folder also includes temperature data for the **1950–1952** period.

As a challenge, repeat the temperature analysis using the 1950–52 data. Calculate the mean temperature associated with each tree observation for both periods and compare the results. Then calculate the difference between the two periods for each observation. Map the resulting `CHANGE` variable and describe the geographic pattern you see.

Challenge: *Which areas of Georgia show the largest changes in the temperatures associated with tree observations? Which trees are in areas with the greatest changes in temperature in this period?*

```
temp1950_to_1952_files<-list.files("data/temp_july1950_1952",
                     pattern=".tif",
                     full.names=T)

temp_1950_to_1952<-rast(temp1950_to_1952_files)
```

```
nlyr(temp_1950_to_1952)
```

```
[1] 93
```

```
temp_1950_to_1952_C<-temp_1950_to_1952 - 273.15

mean(temp_1950_to_1952_C)
```

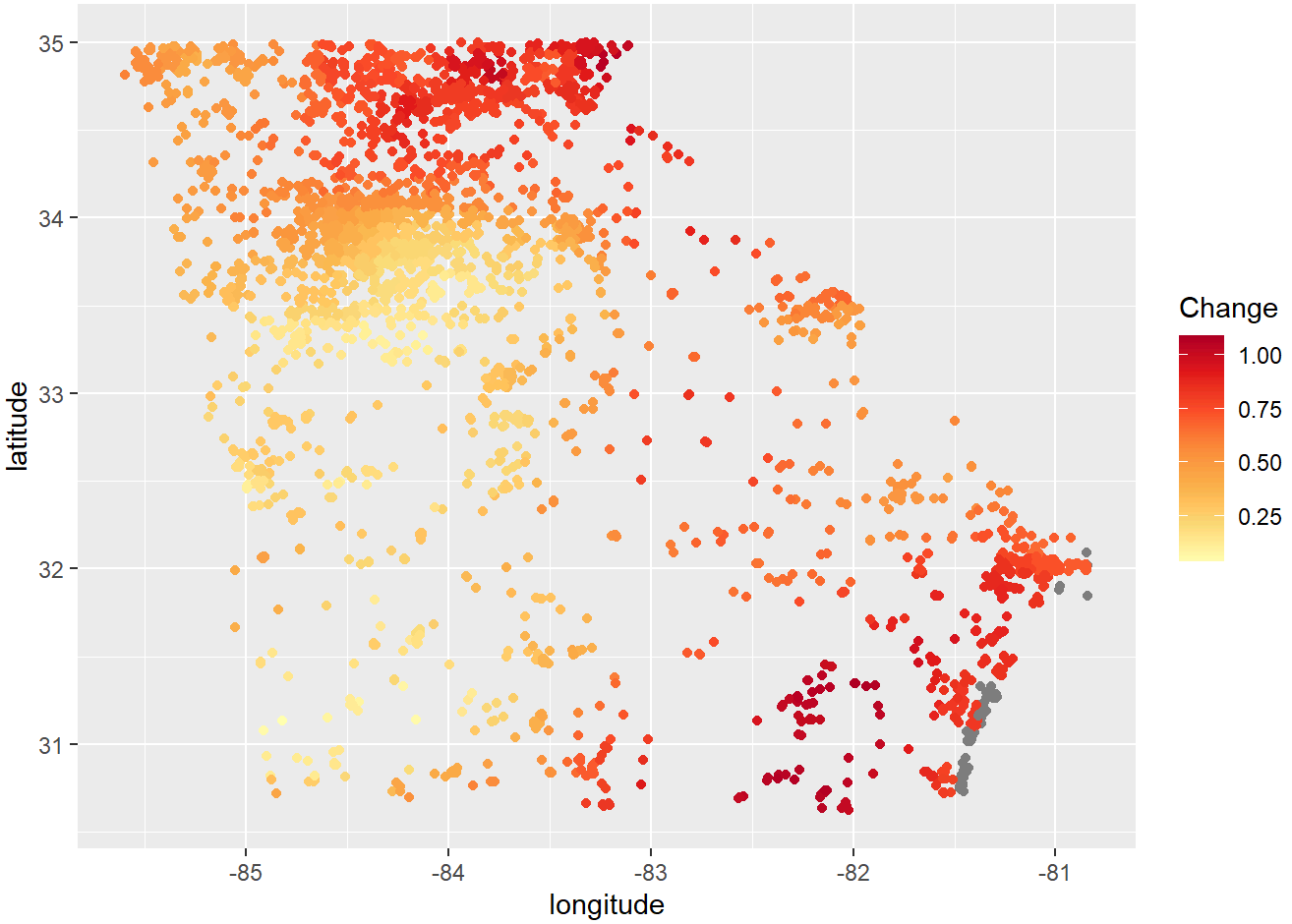
```
class       : SpatRaster
size        : 352, 713, 1  (nrow, ncol, nlyr)
resolution  : 0.08983153, 0.08983153  (x, y)
extent      : -127.0218, -62.9719, 22.99687, 54.61757  (xmin, xmax, ymin, ymax)
coord. ref. : lon/lat WGS 84 (EPSG:4326)
source(s)   : memory
name        :      mean
min value   :  3.006291
max value   : 38.702769
```

```
extracted_tree_temp_data_1950_to_1952_C<-terra::extract(
  x = mean(temp_1950_to_1952_C),
  y = gatree_obs_inat)

gatree_obs_inat$Temperature_1950_to_1952_C<-extracted_tree_temp_data_1950_to_1952_C$mean

gatree_obs_inat$Change<-gatree_obs_inat$Temperature_2022_to_2024_C - gatree_obs_inat$Temperature_

ggplot(gatree_obs_inat)+
  geom_point(aes(x=longitude,
      y=latitude,
      color=Change))+
  scale_color_distiller(palette = "YlOrRd", direction=1)
```



Both the coast and NE Georgia saw almost 1°C increases from 1950-1952 to 2022-2024. This would most directly impact the Live Oak along the Southeastern Georgia coast, and the Rhododendron, which is predominantly in the Northern Georgia area.

Final submission

Disclosure of assistance

Besides class materials, what other sources of assistance did you use while completing this lab? These can include input from classmates, relevant material identified through web searches (e.g., Stack Overflow), or assistance from ChatGPT or other AI tools. How did these sources support your own learning in completing the lab?

I used AI to get a better understanding of tasks, what they were asking for, and how to apply different parts of the data to get the results I wanted. For example, in task 4, I used AI to explain what goes into the Location Quotient equation (LQ=pct_subarea/pctarea). AI helped me understand what kind of data should go in the subarea part of the equation, so I could do the calculation correctly. I also had a classmate explain to me how to do the Location Quotient in R, but primarily used AI for guidance. Also in task 4, I needed help selecting the “Live oak” data since it was 2 words, so AI explained why “” didn’t work, and why I needed to use `` instead. I also used AI to help with task 8 because I was confused about the format for the

terra::extract() function. I had never seen it before so I used AI to see how to format the function to apply it to my own code. I also used it if I got any errors I wasn't sure how to fix. For example, in task 9 I was getting the same temperature, standard deviation, latitude, and longitude value for all 4 species, and AI helped me by telling me I was not letting it filter by the gentype column since I was used the $ to select a column, which would just use the entire column's data instead of just the data in the column that belongs to the specific tree species. I also asked AI how to add a title to the figure in task 6, because I was curious (even though it wasn't necessary to the lab). For the challenge question, I used AI to help me figure out which tool was best to plot the data (I wasn't sure if using plot() or ggplot() would be best), as well as choosing and using the best color scheme. Lastly, I used AI to fix simple errors like forgetting |>, or "" and +, which I forget pretty frequently. These tools helped me complete my lab, as I was stuck several times on different tasks and these tools allowed me to focus learn the formatting and process of coding for this lab and not to get bogged down trying to find the right format or find the missing + or |>.

Lab reflection

How do you feel about the work you did on this lab? Was it easy, moderate, or hard? What are the biggest things you learned by completing it?

Honestly, this one was by far the hardest so far. I was pretty confused with the location quotient task and some of the temperature tasks. I didn't have a great example for some of the tasks to go off of so I had a pretty hard time figuring out the different ways to go about completing a couple tasks. For example, I didn't have any previous example for the terra::extract() function so I had to use AI to understand the format so I could apply it to my task.

The main things I learned are extract .tif file data and how to use R to get statistics from raster data. I also learned how to extract specific data from R, and create a location quotient from two different datasets, joined together (which could be very helpful in my thesis).